STAFF REVIEW · H446 286 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Traffic failing its security rules.

2. Storage retaining content without power.

3. Bit pattern applied with bitwise operators to select/change bits.

4. $00001010_2 = 0A_{16}$ (last four bits 1010 retained).

I Do — source worked steps

1. Predict first — An altered digit can identify a different record and corrupt its meaning.

2. Model and check — Lossless encoding is essential; compression ratio depends on redundancy in the source.

We Do — guided CHECK

1. Predict first
A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.
Model and check
Keep originals separately so discarded detail can be recovered for future editing.

You Do — source mark schemes

1. Lossless exact reconstruction [2], use source/text/records [1]; lossy discards information [2], use delivery media [1].

2. (A,3),(B,2),(A,1) [3]; three A [1], two B [1], one A [1].

3. Audio lossy if acceptable sound/size [2]; exam marks lossless [2]; lossy removes exact detail [1], preservation of originals/recovery [1].

Exit Ticket — source answer

1. False. Some data must remain exact.